Documentation to the WBQV software

WAC consortium

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Chapter 1

Frequency response

1.1 Equations

Definitions:

- Digitizer: measures amplitude $D(f)$, have got frequency dependent gain $G(f)$. Time dependence of the gain is neglected.
- Thermal transfer standard: U is output DC voltage measured by the multimeter, and the overall gain is $\Delta(f)$. Thermal transfer standard and the multimeter are supposed to be stable in time.
- AC source: generates signal of amplitude with non-negligible time and frequency dependence $Z(f,t)$.

One measurement of the AC source by digitizer and by the thermal transfer standard is expressed as:

$$D(f,t) = Z(f,t) \cdot G(f) \tag{1.1}$$
$$U(f,t) = Z(f,t) \cdot \Delta(f) \tag{1.2}$$

If the measurement by digitizer and by the themral transfer standard is done simultaneously, or almost simultaneously, then:

$$D(f) = Z(f) \cdot G(f) \tag{1.3}$$
$$U(f) = Z(f) \cdot \Delta(f) \tag{1.4}$$

Two measurements are done, for measured and reference frequencies marked as f_{M} and f_{R}. If these measurements are done right after another,

then the time dependence of the AC source can be neglected and only frequency dependence is taken into account:

$$t_{\mathrm{R}} \approx t_{\mathrm{M}} \implies Z\big|_{t_{\mathrm{R}}} = Z\big|_{t_{\mathrm{M}}} \tag{1.5}$$

Then we can write both measurements as:

$$D(f_{\mathrm{M}}) = Z(f_{\mathrm{M}}) \cdot G(f_{\mathrm{M}}) \tag{1.6}$$
$$U(f_{\mathrm{M}}) = Z(f_{\mathrm{M}}) \cdot \Delta(f_{\mathrm{M}}) \tag{1.7}$$
$$D(f_{\mathrm{R}}) = Z(f_{\mathrm{R}}) \cdot G(f_{\mathrm{R}}) \tag{1.8}$$
$$U(f_{\mathrm{R}}) = Z(f_{\mathrm{R}}) \cdot \Delta(f_{\mathrm{R}}) \tag{1.9}$$

Combining the equations:

$$\frac{G(f_{\mathrm{M}})}{G(f_{\mathrm{R}})} = \frac{D(f_{\mathrm{M}})}{D(f_{\mathrm{R}})} \cdot \frac{U(f_{\mathrm{R}})}{U(f_{\mathrm{M}})} \cdot \frac{\Delta(f_{\mathrm{M}})}{\Delta(f_{\mathrm{R}})} \tag{1.10}$$

For f_{M} varied through whole input bandwidth of the digitizer, the ratio returns the frequency response of the digitizer related to the gain at reference frequency f_{R}. If the reference frequency is small enough, the gain $G_{f_{\mathrm{R}}}$ can be measured using DC method.

DC offset is not result of measurement and processing!

AC-DC measurements:

$$\Delta_L = 1 + \delta(f) \tag{1.11}$$
$$U_{\mathrm{AC_{in}}} = U_{\mathrm{DC_{in}}} \cdot (1 + \delta \cdot 10^{-1}) \tag{1.12}$$

With:

$$\delta_{792} = \frac{U_{\mathrm{out_{AC}}} - U_{\mathrm{out_{DC}}}}{n \cdot U_{\mathrm{out_{DC}}}} \tag{1.13}$$

1.2 Interpolation of the FR function

The FR of the National Instruments 5922 digitizer is of a complex shape. For calculation of SS data, the interpolation of the measured FR data is needed, because the calibration of FR can happen at different frequencies than required for SS measurement.

The FR shape can be interpolated by using polynomial interpolation, however the FR response would require polynomials of a high degree. Such

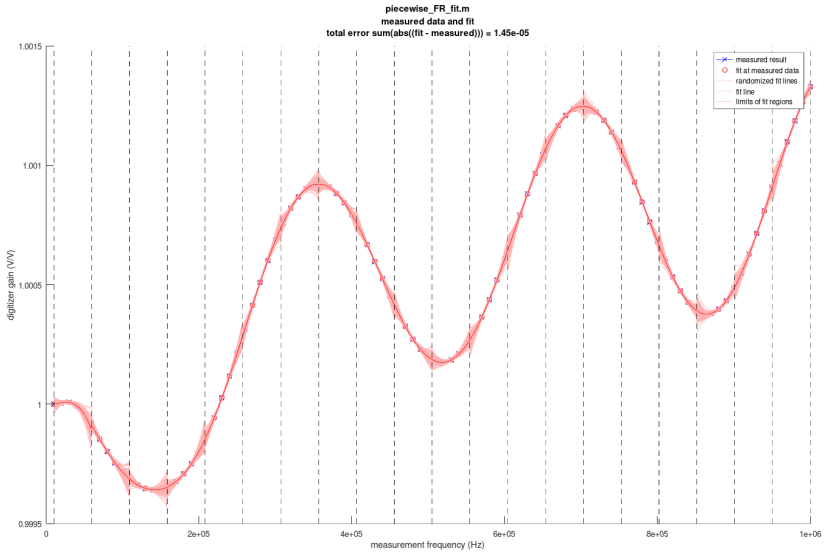


Figure 1.1: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

polynomials tends to oscillate between the measured points and especially at leftmost and rightmost parts of the function. The leftmost part of the function is at lower frequency and has the highest impact on the rms value obtained from SS data, thus such interpolation is not suitable.

The possibility is to use piece-wise polynomial interpolation, that mitigates the issue of using high degree polynomials. Naive use of piece-wise polynomial interpolation (just splitting into pieces) does not guarantee smoothness of the whole interpolated function at the edges of the pieces. Later the interpolated curve has to be used also during the Monte Carlo method. Typically when fitting data, the measured points are randomized, multiple fits are estimated, and mean and unceratinty of the fit coefficients are calculated. Later the coefficients are randomized according to the estimated uncertainty, and the uncertainty is propagated through the fit. Randomizing the coefficients of piece-wise polynomials independently cause large discontinuities at the edges of the pieces, see figures 1.1 1.2 and 1.3. To prevent the errors, the randomization of the fit coefficients would need to take into account the requirement of smoothness at the boundaries of the pieces.

GNU Octave function `splinefit` provides simple way to use piece-wise spline (third order polynomials) interpolation, and assures smoothness at the

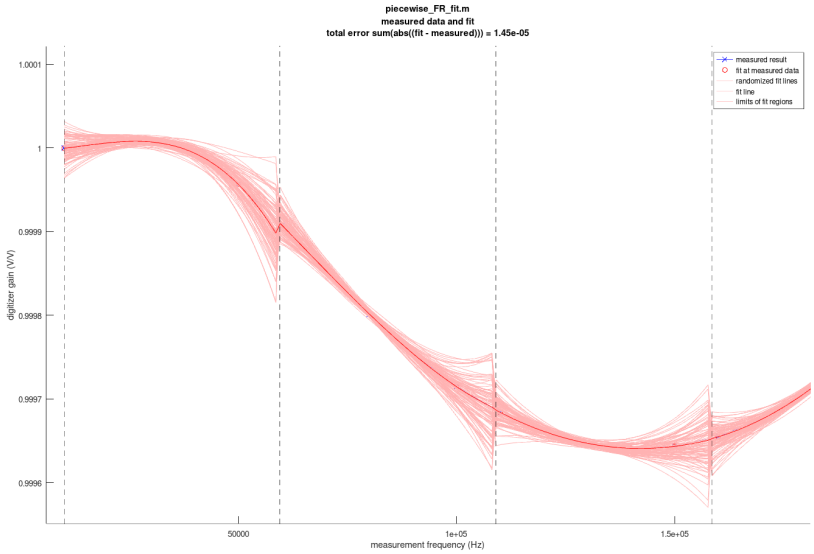


Figure 1.2: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

boundaries of the pieces. However still the coefficients cannot be randomized independently. To circumvent the issue, following approach is used. The FR measurement data are randomized and multiple spline piece-wise fits are estimated. Coefficients of the all fits are saved into an intermediate file. The intermediate file has to consist of sufficient number of fits to provide good statistics for the uncertainty propagation using Monte Carlo method. Later, when uncertainty has to be propagated throught the spline fit, coefficients are loaded and a random selection of the coefficients provides the Monte Carlo method propagation of uncertainties throught the fit.

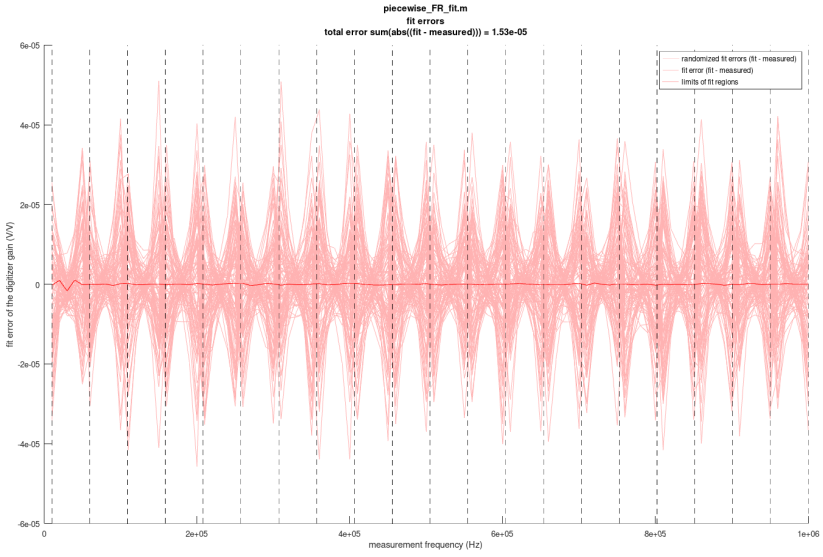


Figure 1.3: Error of the piece-wise fit compared to the measurement data. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

Chapter 2

Graphical User Interface

2.1 Main window

After starting the GUI, window (fig 2.1) appears. It can be used to show other windows, or to show status of devices and status of the running measurement.

2.1.1 What to do here

First setup TWM, AC source, DC meter, PJVS, than FR measurement, run FR measurement, setup CE measurement, run CE measurement, setup DG measurement, run DG measurement, SS measurement, run SS measurement.

2.1.2 Elements of the window

- Button `About` – shows a new window with informations on the software.
- Button `Setup TWM & proc` – shows a new window with settings for `TWM` and `GNU Octave`.
- Button `Setup AC Source` – shows a new window with settings for AC source (sec. 2.2). The button `Manual control` right below shows a new independent window for direct manual control of an AC source (sec. 2.2).
- Button `Setup DC meter` – shows a new window with settings for DC meter. The button `Manual control` right below shows a new independent window for direct manual control of a DC meter.
- Button `Setup PJVS` – shows a new window with settings for PJVS. The button `Manual control` right below shows a new independent window for direct manual control of a PJVS.

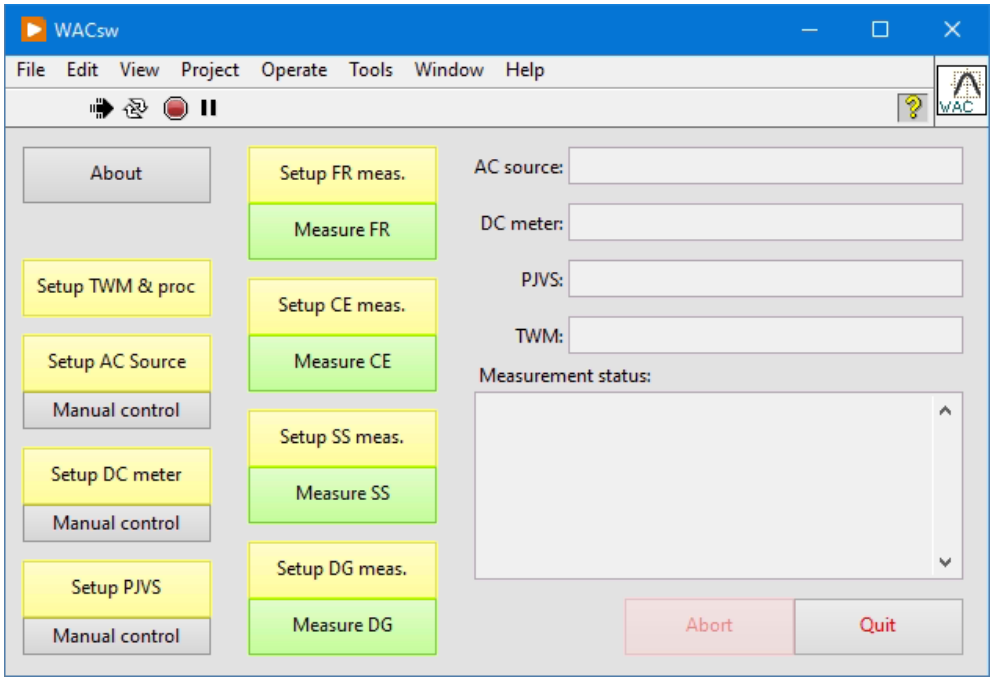


Figure 2.1: Main GUI window.

2.2 AC Source

2.2.1 Setup AC source

The AC source is used for FR, CE and SS measurements. The setup is done using window 2.2 (started from 2.1).

2.2.2 What to do here

Setup AC source (type, VISA address), if the AC source needs to be referenced by AC signal (e.g. FLuke calibrators), use checkbox `Synchronization by AC signal` and setup source for the synchronization signal. Check the settings by using button `Test connection`.

2.2.3 Elements of the window

- Input `AC source type` –
- Input `VISA` –

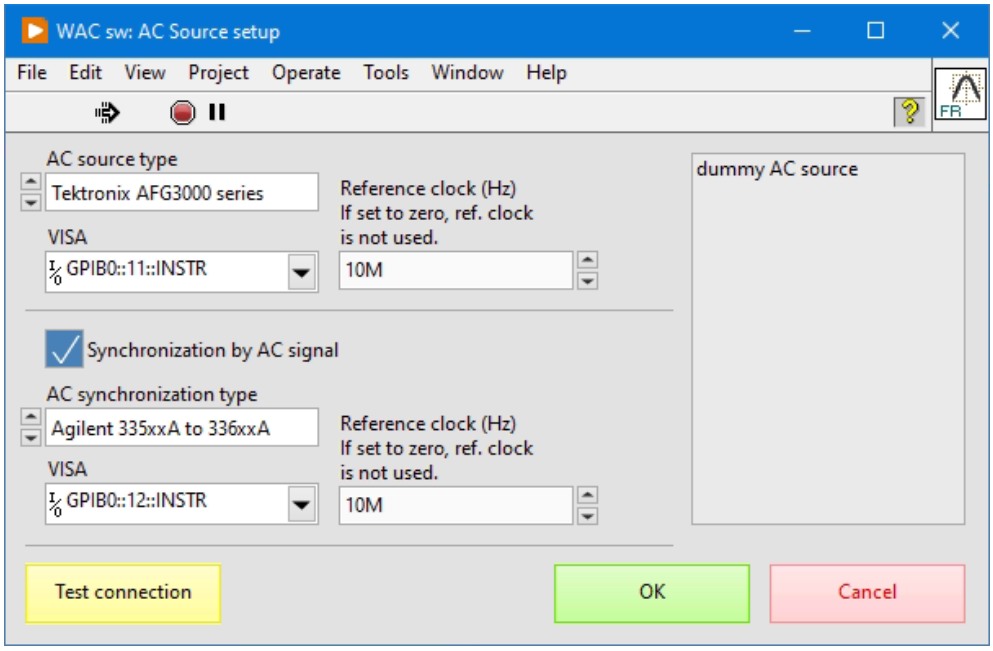


Figure 2.2: Window to setup AC source.

- Input `Reference clock` –
- Checkbox `Synchronization by AC signal` –
- Button `Test connection` –
- Button `OK` –
- Button `Cancel` –

2.2.4 Manual control of AC source

The AC source can be controlled manually using window 2.3.

What to do here: setup your AC source on the top left (if needed), if `Synchronization by AC signal` is selected, setup also AC source on the top right (checkbox `Synchronization by AC signal` in the right one has no effect). Set required signal frequency, amplitude, ensure timeout is long enough. Press `Initialize`, `Start`. Now AC source should be generating signal. Press `Stop` to stop generation.

Elements of the window:

- Element group for AC source –
- Element group for AC source used for Synchronization –
- Input `Signal frequency` –
- Input `Signal amplitude` –
- Input `General settings: Timeout value` –
- Box `Status` –
- Button `Initialize` –
- Button `Start` –
- Button `Stop` –

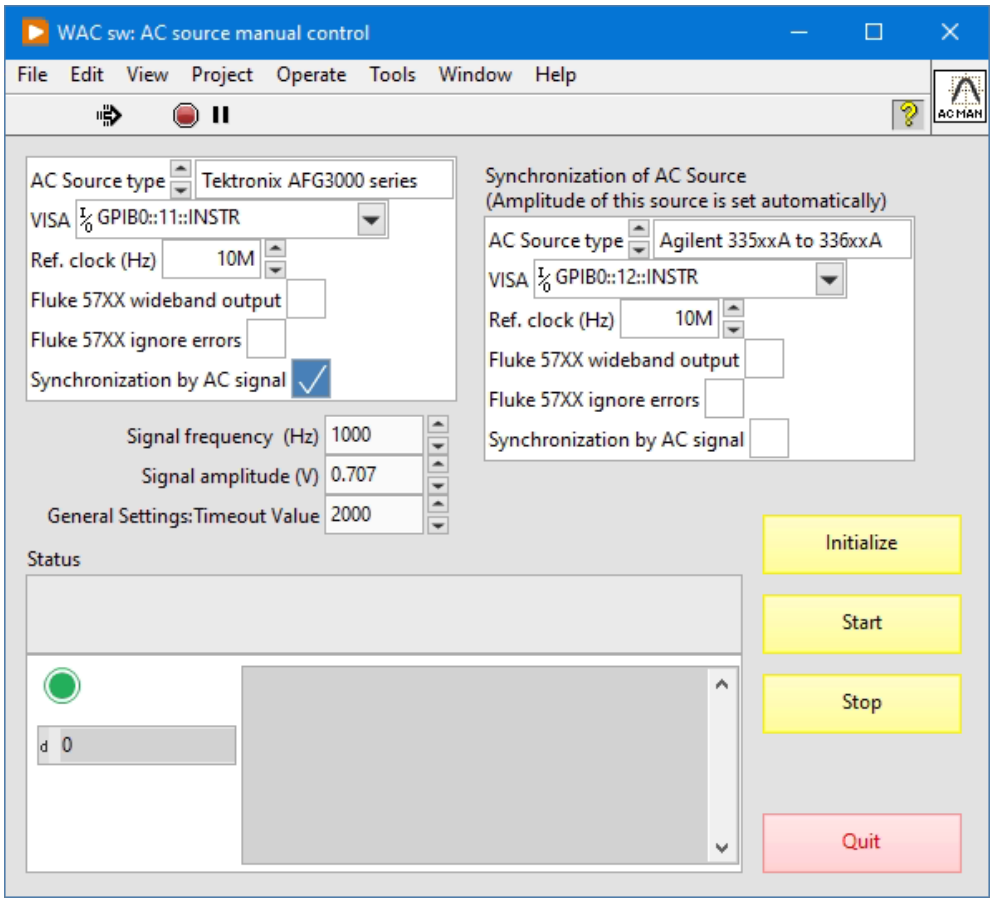

WAC sw: AC source manual control
File Edit View Project Operate Tools Window Help
AC MAN
AC Source type
Tektronix AFG3000 series
VISA
GPIB0::11::INSTR
Ref. clock (Hz)
10M
Fluke 57XX wideband output
Fluke 57XX ignore errors
Synchronization by AC signal
Synchronization of AC Source
(Amplitude of this source is set automatically)
AC Source type
Agilent 335xxA to 336xxA
VISA
GPIB0::12::INSTR
Ref. clock (Hz)
10M
Fluke 57XX wideband output
Fluke 57XX ignore errors
Synchronization by AC signal
Signal frequency (Hz)
1000
Signal amplitude (V)
0.707
General Settings:Timeout Value
2000
Status
d 0
Initialize
Start
Stop
Quit

Figure 2.3:

XXXXXXXXXXXXXXXXXXXXX

	A	B	C	D	E	F	G	
4	*Results will be written into collumns D-H (and K+). Sheet must*			AC source settle time	1		Digitizer idn	NI PXI-592
5	*contain keywords* ***START*** *and* ***STOP*** *in collumn A, values of*			PJVS cable length	5.1			
6	*frequency in collumn B, and record length in C.*			PJVS cable length unc.	0.1	m		
7	***Place keyword STOP to the row with half of the sampling frequency.***			Short cable length	0.1	m		
8				Short cable length unc.	0.01	m		
9				Algorithm	TWM-WRMS	m		
10				Path to FR fit	Example of FF meter template 4 MHz_fit.mat			
11	*inputs*				*results*			
12	Command	Signal frequency	Record length	Switch position	TIme of reading	digitizer amplitude rms		
13	(START, STOP)	Hz	multiple of periods	0 or 1	(time of writing t	value (V)	unc (V)	
14	START	1 000	100	1	28.11.2025 12:07	1.000	0.000001	
15		1 000	100	0	28.11.2025 12:07	1.000	0.000001	
16		1 200	120	1	28.11.2025 12:07	1.000	0.000001	
17		1 200	120	0	28.11.2025 12:07	1.000	0.000001	
18		1 350	135	1	28.11.2025 12:08	1.000	0.000001	
19		1 350	135	0	28.11.2025 12:08	1.000	0.000001	
20		1 600	160	1	28.11.2025 12:08	1.000	0.000001	
21	STOP	1 600	160	0	28.11.2025 12:08	1.000	0.000001	

Figure 2.4: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Frequency response measurement template**			*measurement properties*			*hardware data*				Actual number of readings		
2	*Software will read settings from cells E2 to E7 and*			AC voltage amplitude	0.7	V	AC source id	Fluke 5XXX s.n. 689			Crude estimate of measure		
3	*measurement points from collumns A-C (yellow cells).*			Sampling frequency	4 000 000	Hz	DC meter id	Keithley 2182					
4	*Results will be written into collumns D-H (and K+). Sheet must*			AC-DC settle time	20	s	Digitizer idn	NI PXI-5922, sn. 204B7F6, channel 0					
5	*contain keywords* ***START*** *and* ***STOP*** *in collumn A, values of*			AC-DC warm up time	900	s							
6	*frequency in collumn B, and record length in C.*			DC meter readings	5								
7	***Place keyword STOP to the row with half of the sampling frequency.***			Algorithm	TWM-WRMS								
8				Simultaneous meas.?	1								
9													
10													
11	*inputs*			*results*									
12	Command	Signal frequency	Record length	TIme of DC readings	Digitizer amplitude rms		DC voltage				DC meter readings		
13	(START, STOP)	Hz	multiple of periods	(time of writing this line i	value (V)	unc (V)	value (V)	unc (V)			|---------->		
14	START	10000	10000	29.06.2026 11:47	0.888857	0.000000	0.772344	0.000000	10	1.00	0.772	0.772	0
15		1000	1000	29.06.2026 11:47	0.888839	0.000000	0.772330	0.000001	###	###	0.772	0.772	0

Figure 2.5: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

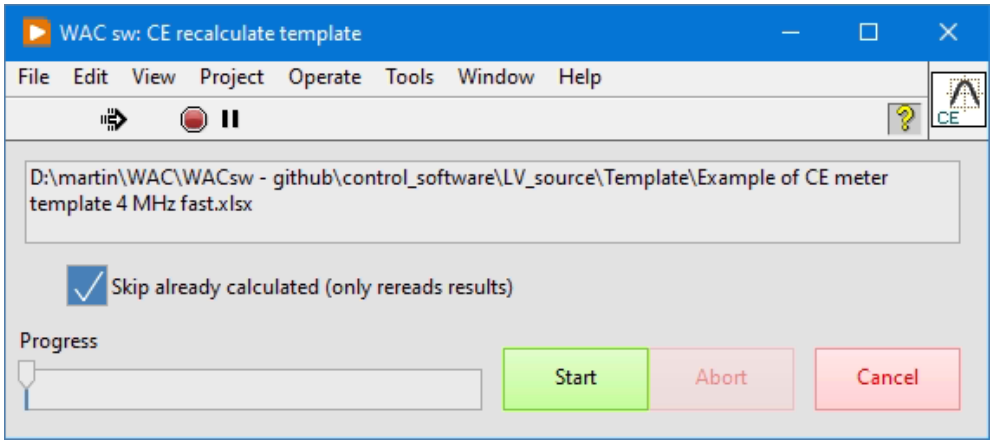


Figure 2.6: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

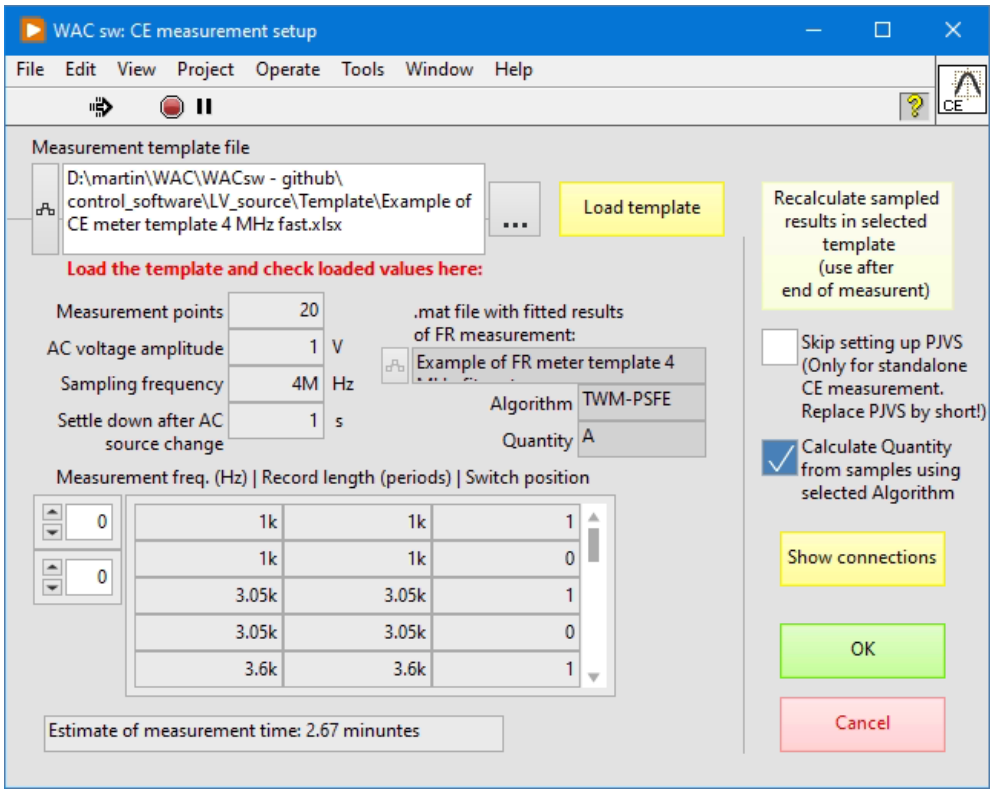


Figure 2.7: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

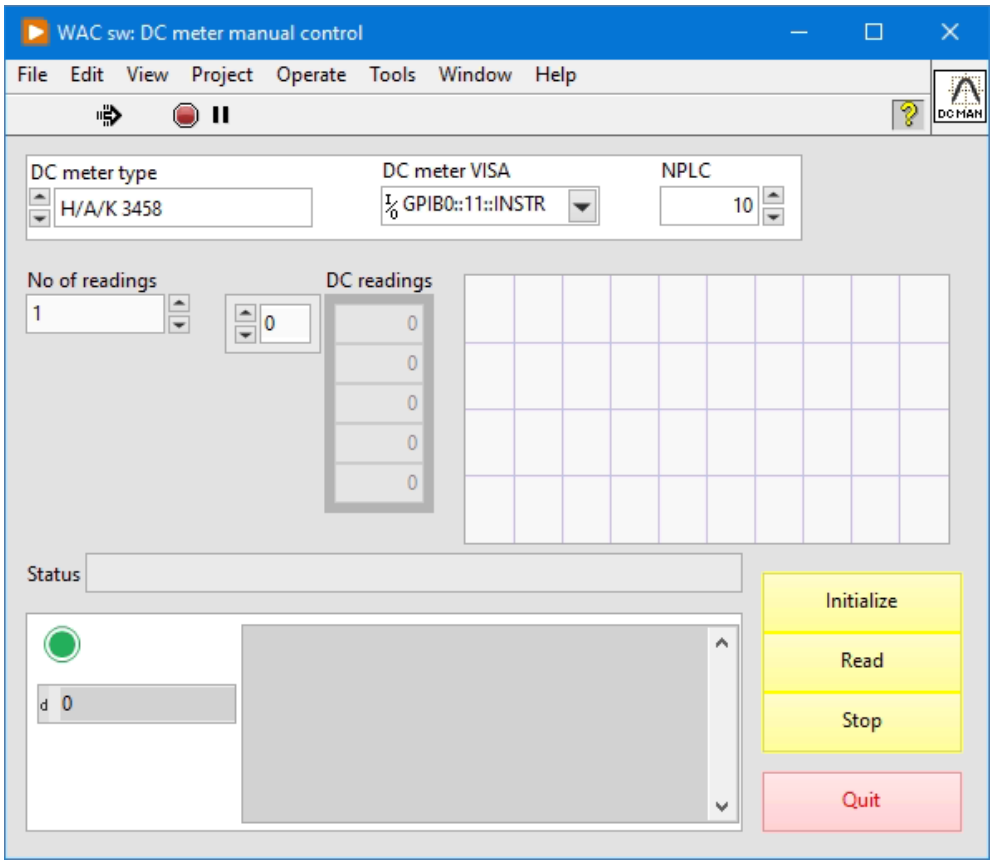


Figure 2.8: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

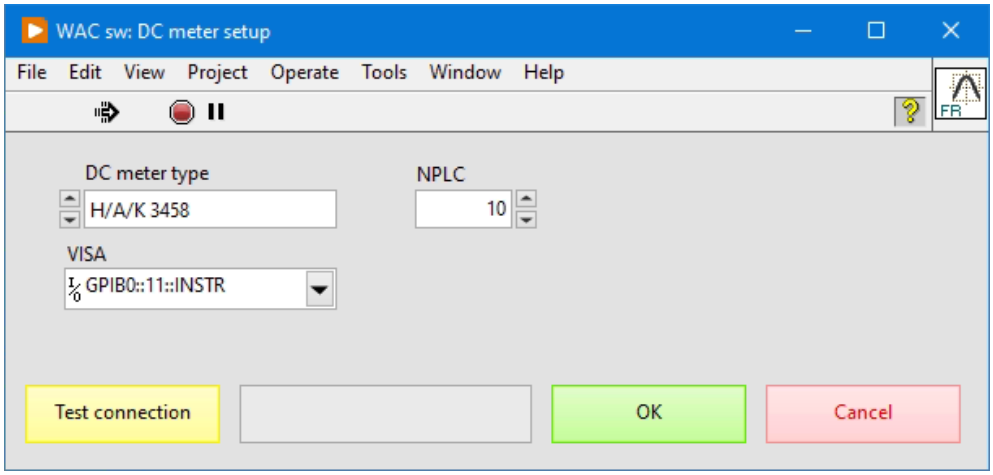


Figure 2.9: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

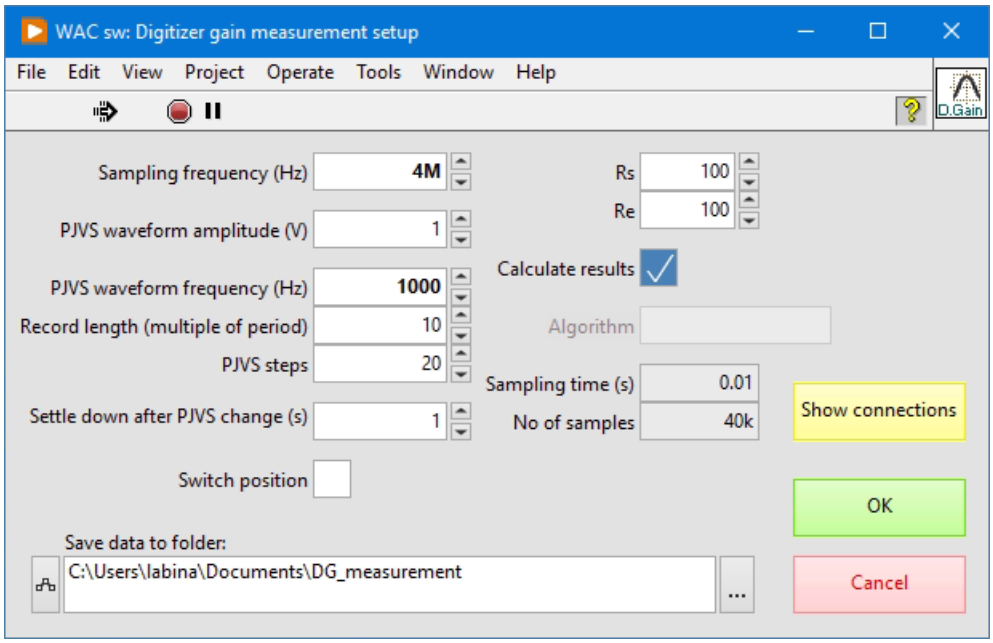


Figure 2.10: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

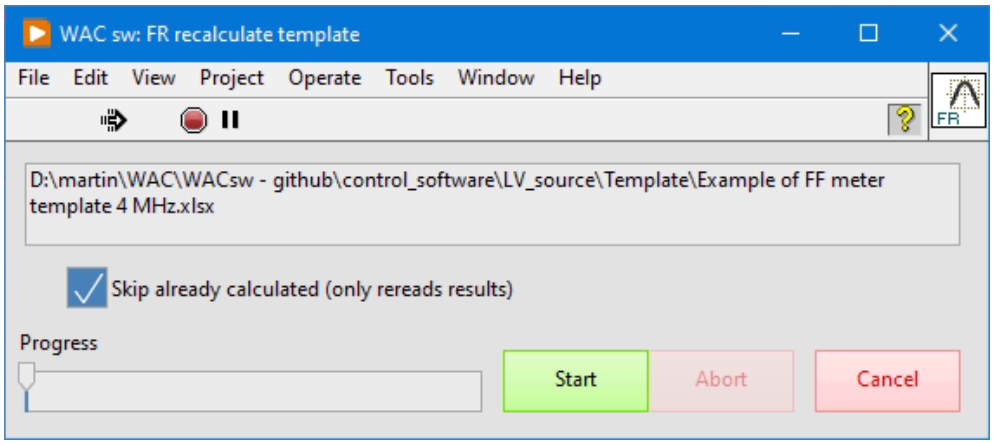


Figure 2.11: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

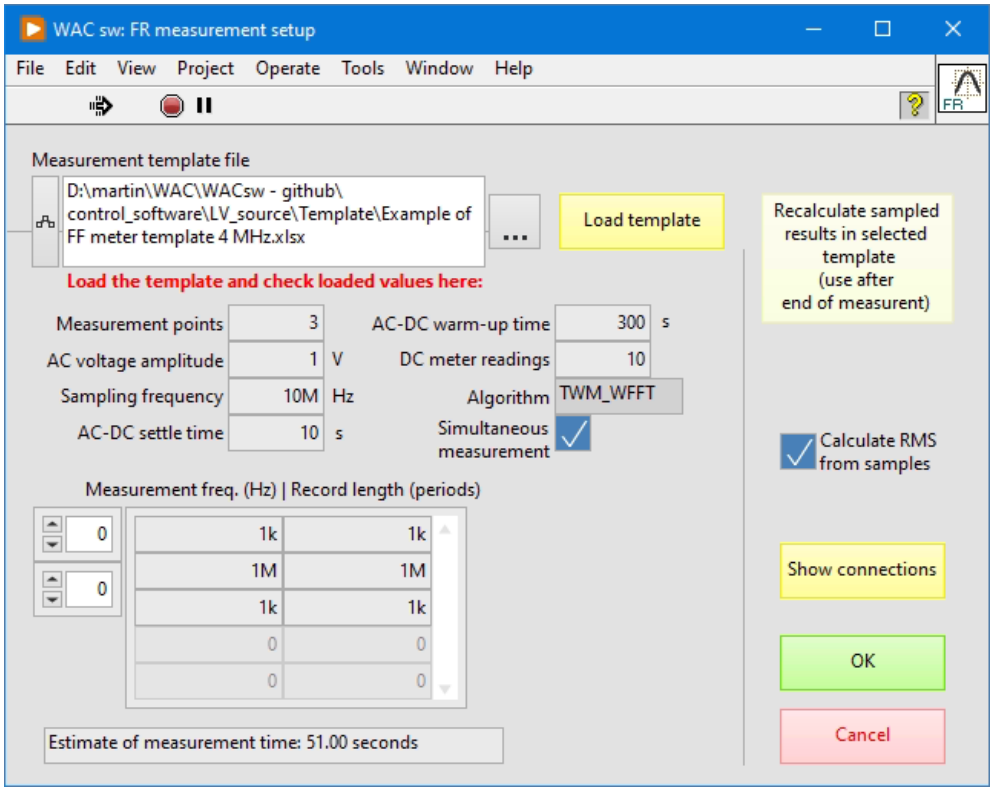


Figure 2.12: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

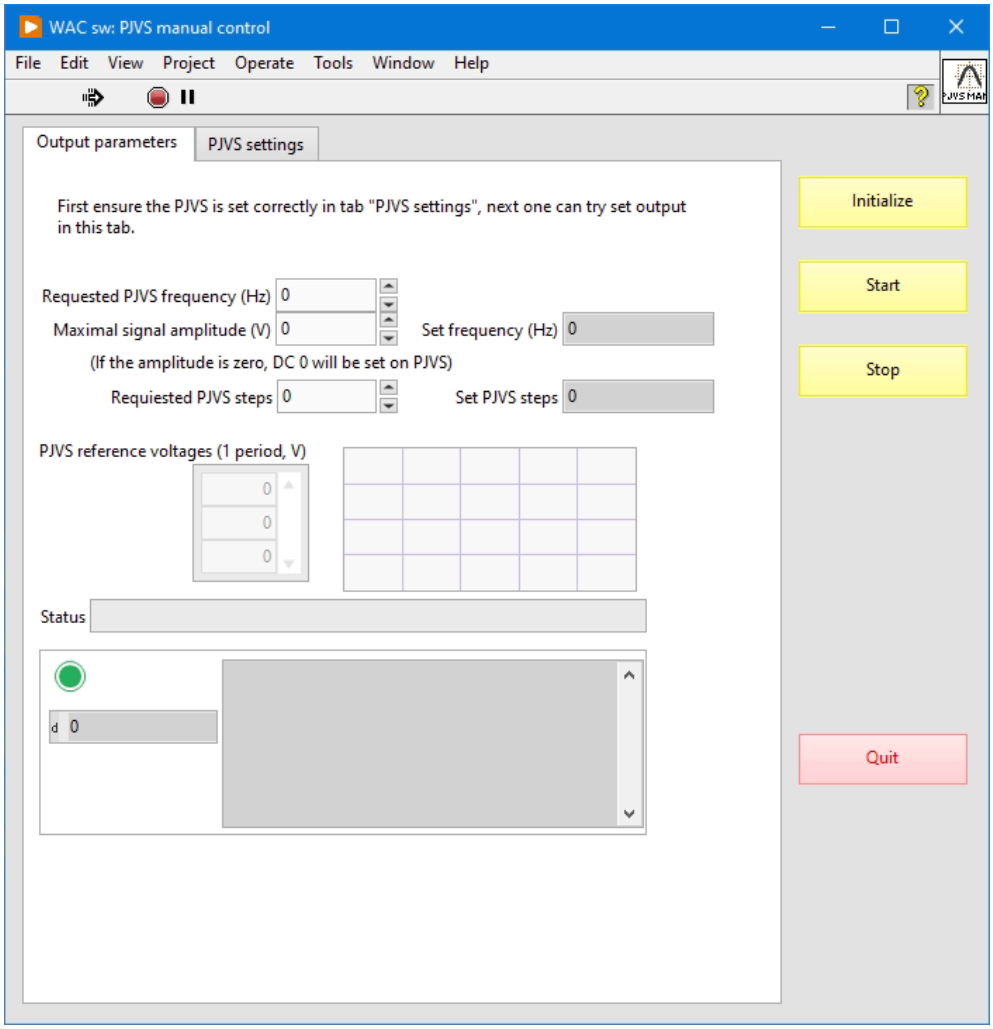


Figure 2.13: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

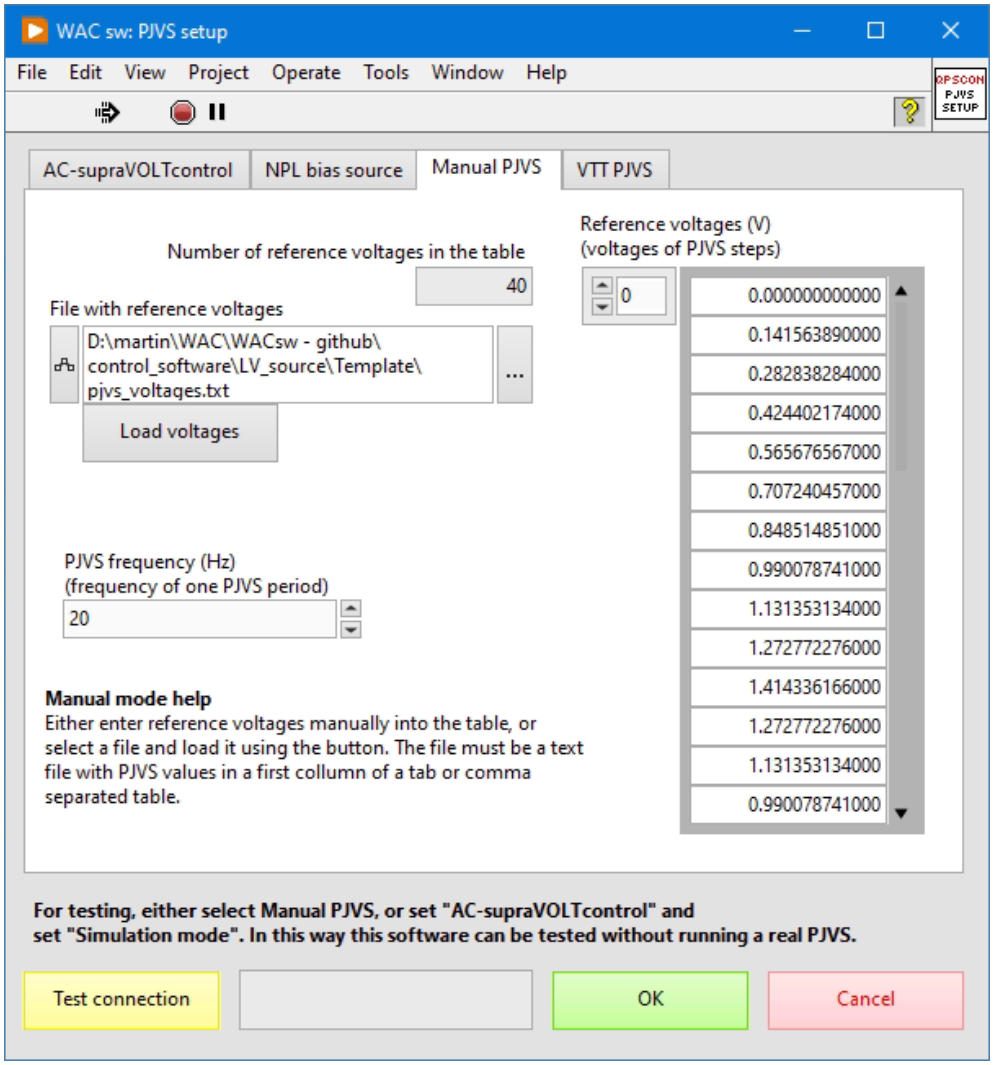


Figure 2.14: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

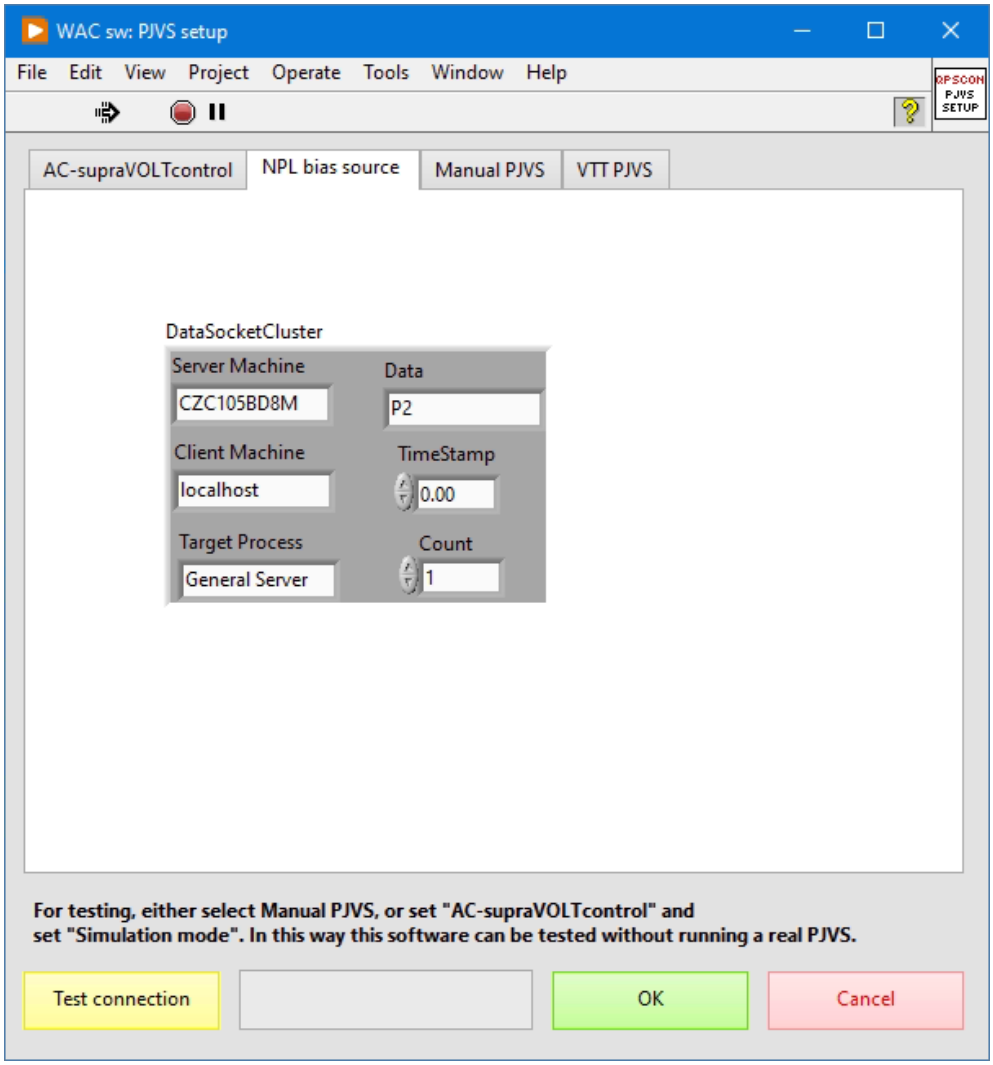


Figure 2.15: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

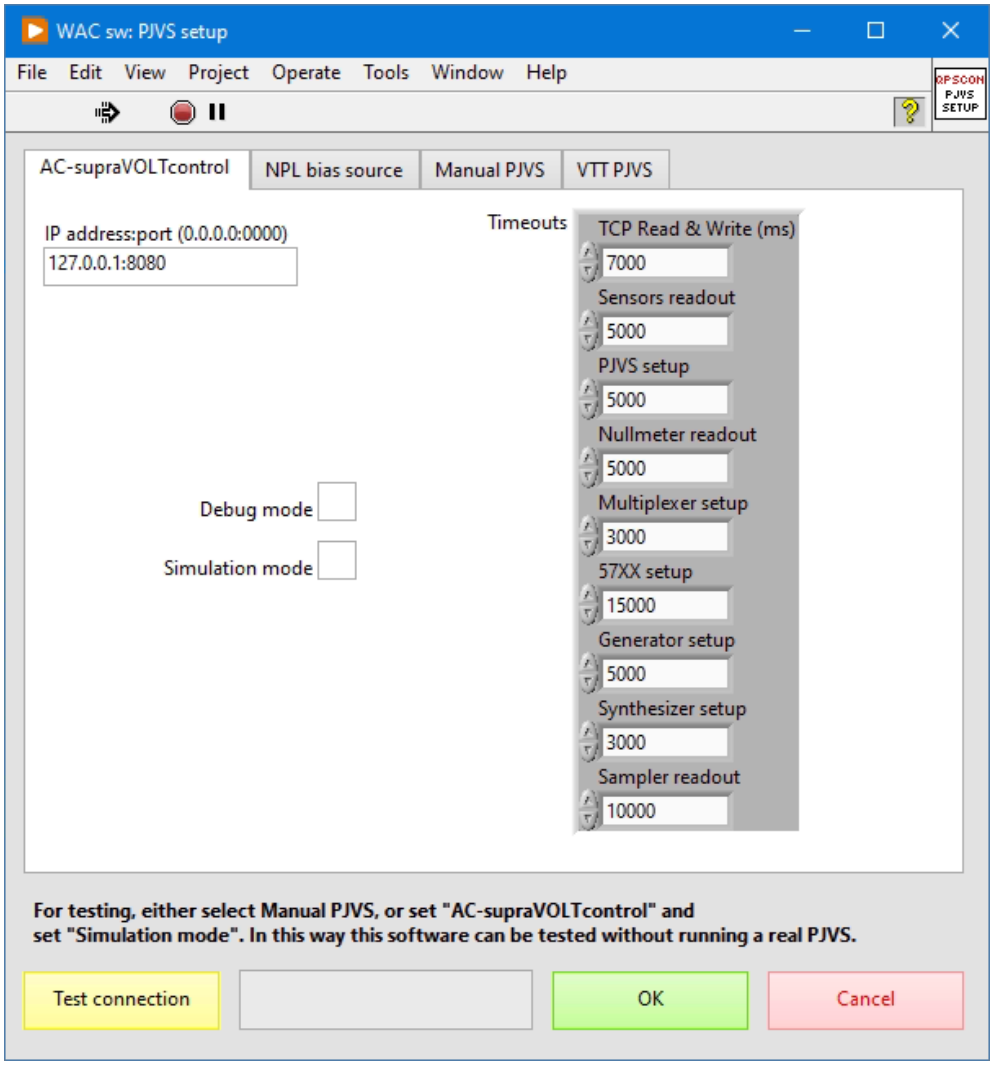


Figure 2.16: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

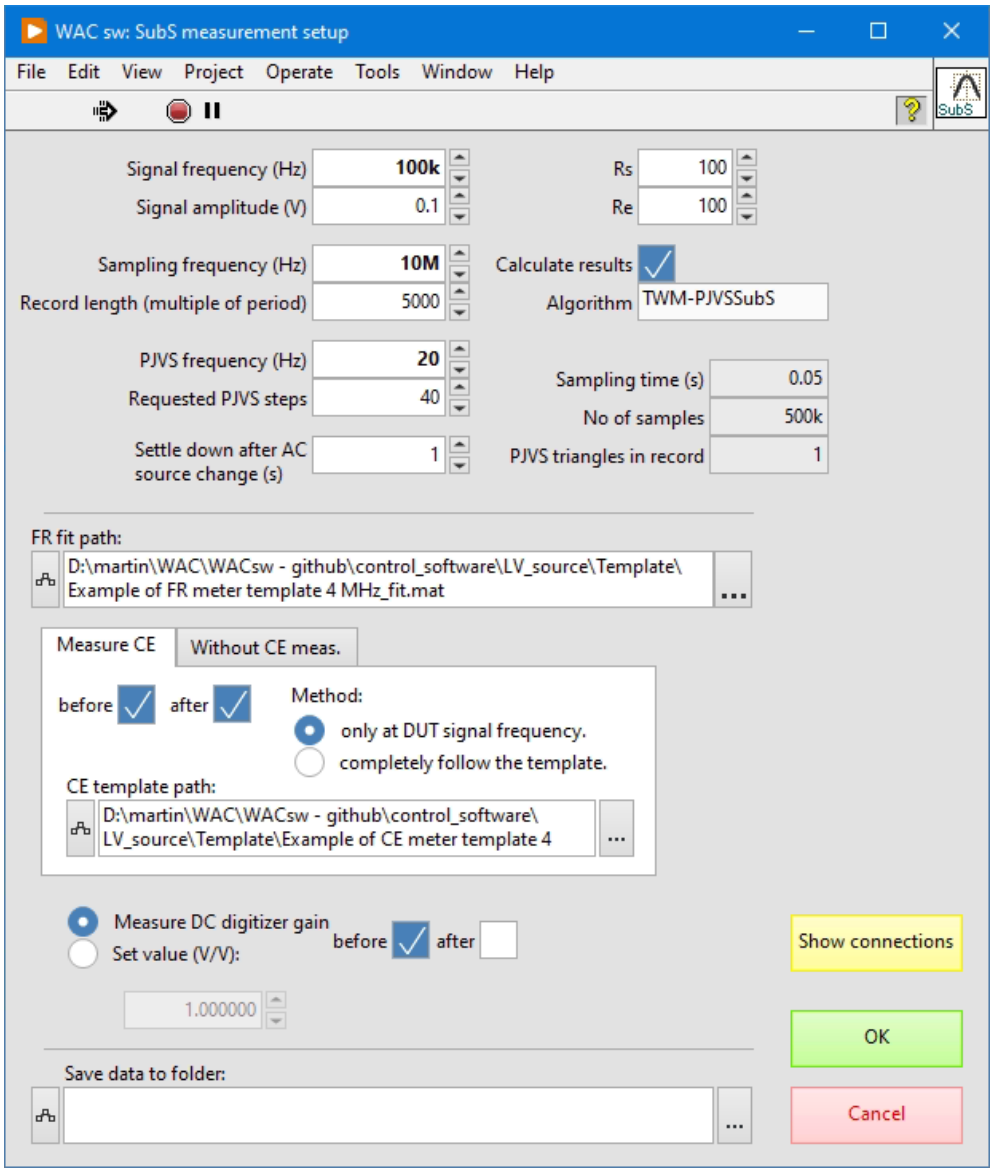


Figure 2.17: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

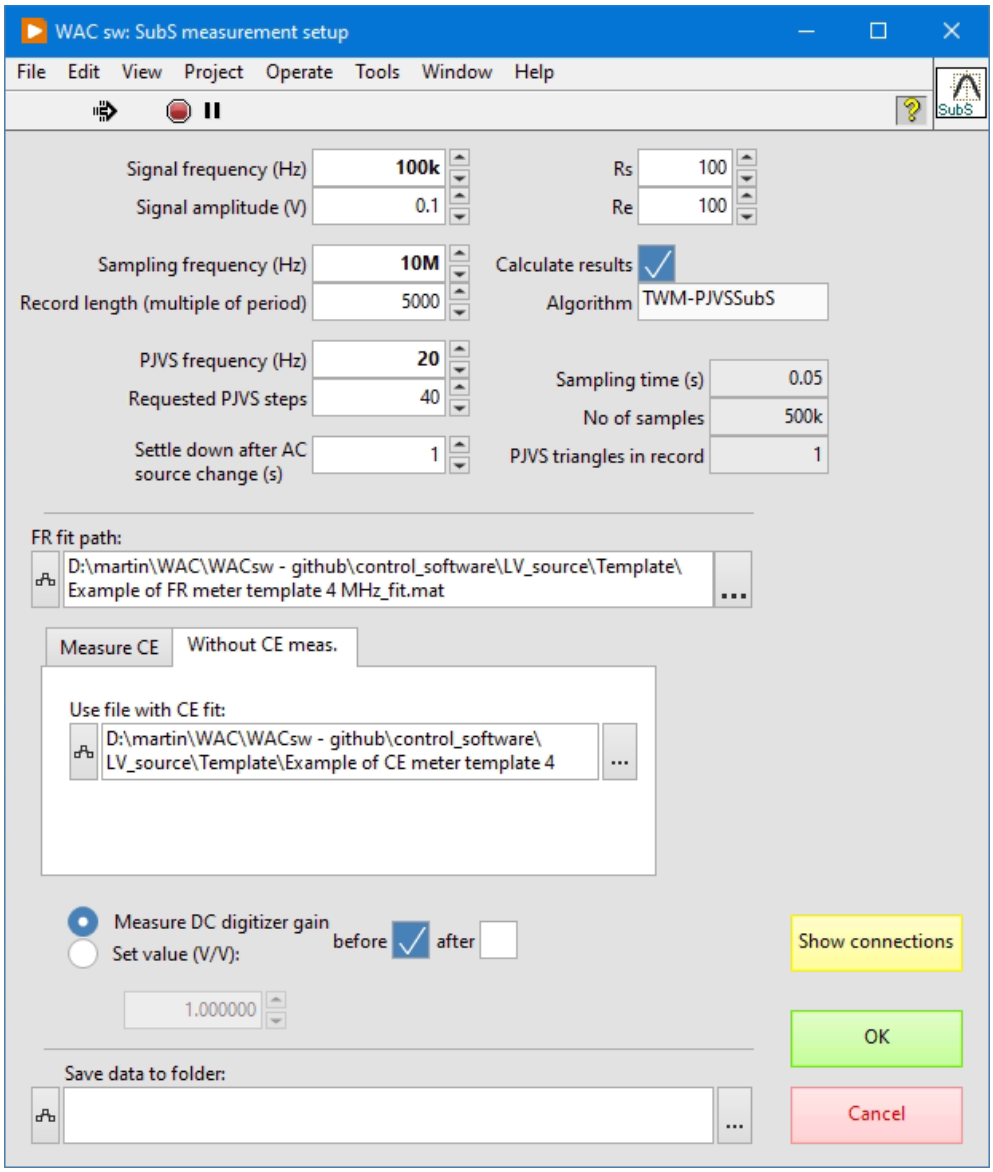


Figure 2.18: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

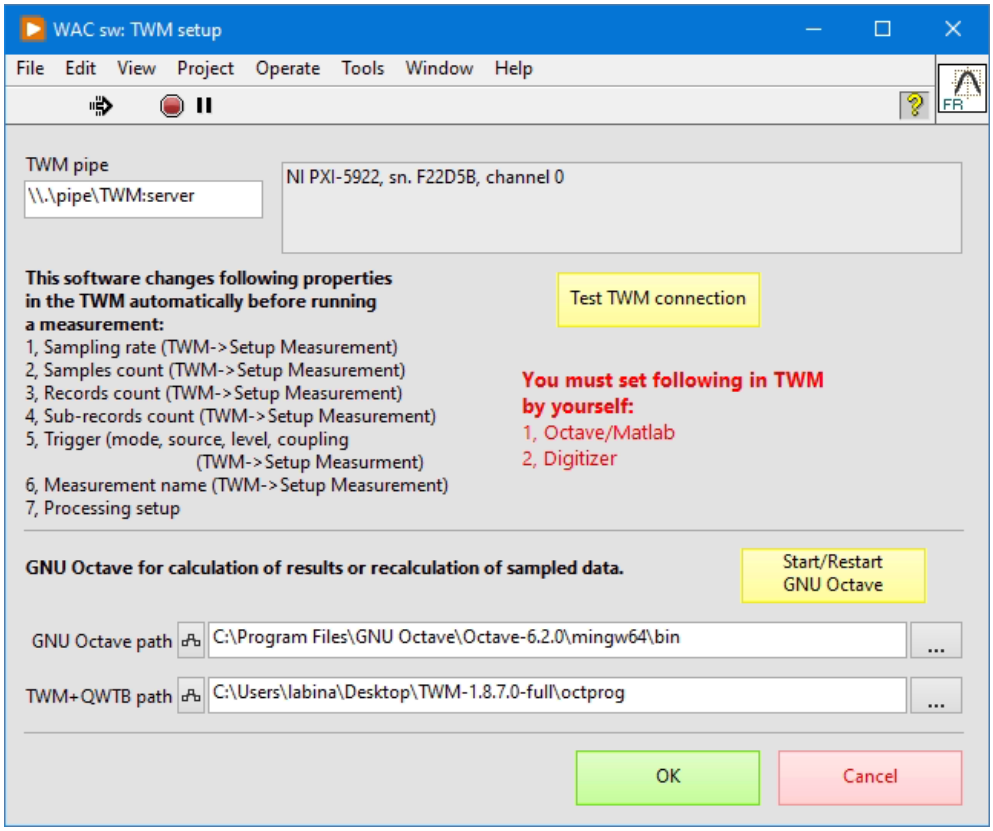


Figure 2.19: Piece-wise fit of a part of frequency response of a digitizer. The ligth red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.